

IS2110 - Data Structures and Algorithms II

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Course Breakdown

- 15 Lecture Sessions
- Assignments - 30%
- Final Exam – 70%

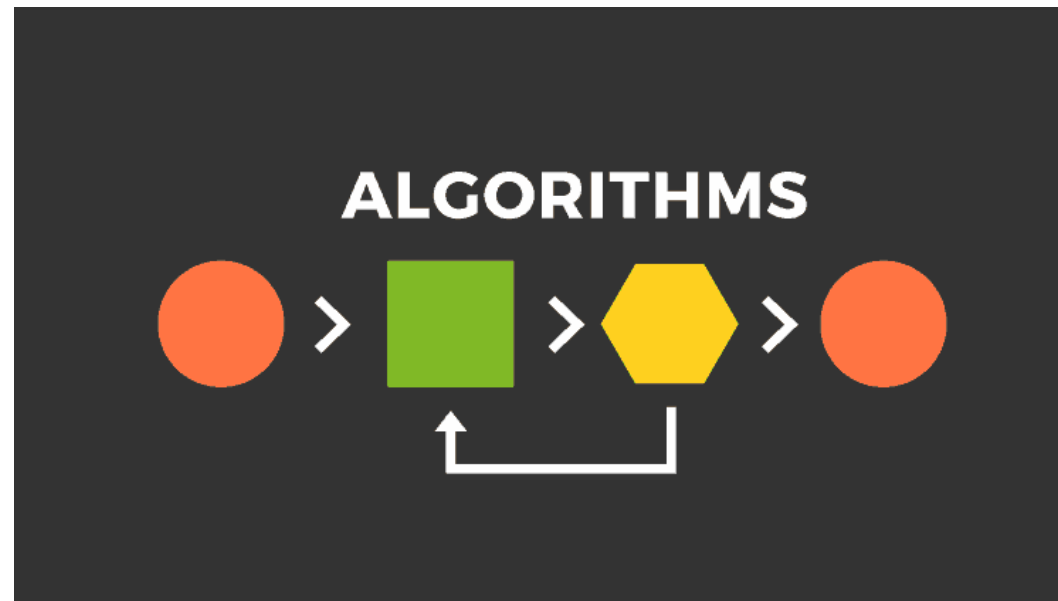
Data structures and algorithms

Why this subject is important?

Algorithms

What is an algorithm-***algorithm*** is any well-defined computational procedure that takes some value, or set of values, as ***input*** and produces some value, or set of values, as ***output***.

tool for solving a well-specified ***computational problem***



Algorithms - Different Aspects

1. Correctness of an algorithm
2. Different solutions to the same problem
3. Complexity
 - Time Complexity
 - Space Complexity

Lesson 1 - Graph Basic Concepts

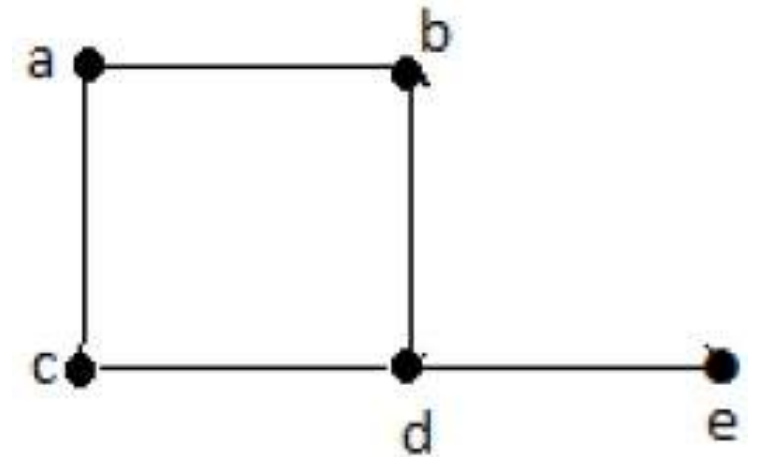
The following slides are based on the sources obtained from the below Tutorial
https://www.tutorialspoint.com/graph_theory/graph_theory_fundamentals.htm

What is a graph

A graph is a pictorial representation of a set of objects where some pairs of objects are connected by links

The interconnected objects are represented by points termed as **vertices**, and the links that connect the vertices are called **edges**.

Formally, a graph is a pair of sets **(V, E)**, where **V** is the set of vertices and **E** is the set of edges, connecting the pairs of vertices



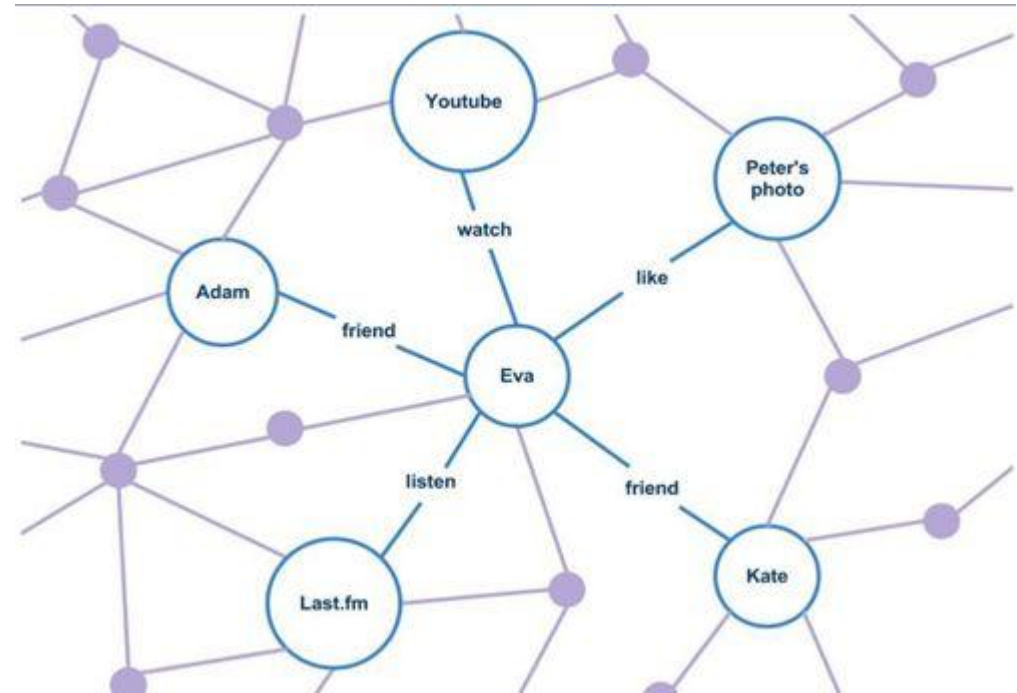
In the above graph,

$V = \{a, b, c, d, e\}$

$E = \{ab, ac, bd, cd, de\}$

Applications of Graph Theory

- Computer Science
- Electrical Engineering
- Computer Network
- Science
- General LIFE



Graph Fundamentals



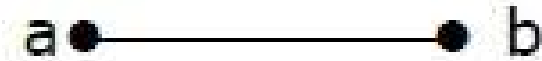
Point

- **A point** is a particular position in a one-dimensional, two-dimensional, three-dimensional , or n-dimensional space.
- For better understanding, a point can be denoted by an alphabet. It can be represented with a dot.



Line

- A **Line** is a connection between two points. It can be represented with a solid line.
- Here, 'a' and 'b' are the points. The link between these two points is called a line.



Vertex

- A vertex is a point where multiple lines meet. It is also called a **node**.
- Similar to points, a vertex is also denoted by an alphabet.



Edge

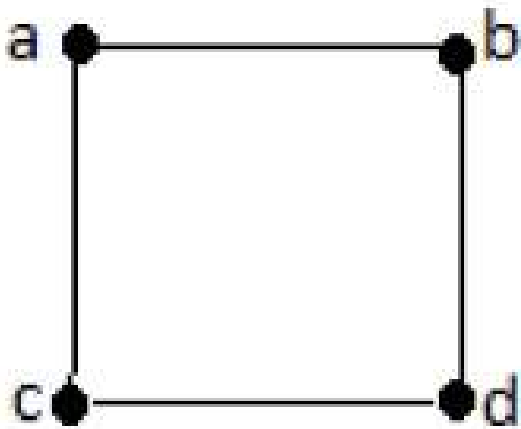
- An edge is the mathematical term for a line that connects two vertices (nodes).
- Many edges can be formed from a single vertex.
- Without a vertex, an edge cannot be formed. There must be a starting vertex and an ending vertex for an edge.



Here, 'a' and 'b' are the two vertices and the link between them is called an edge.

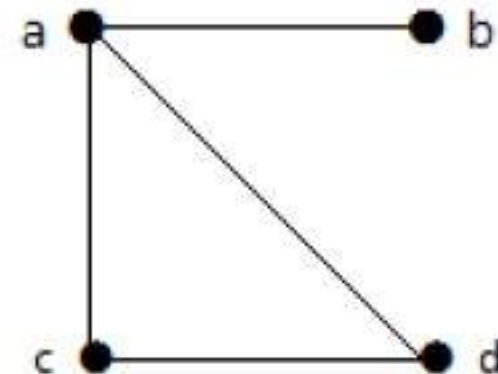
Graph

A graph 'G' is defined as $G = (V, E)$ Where V is a set of all vertices and E is a set of all edges in the graph.



$V = \{a, b, c, d\}$

$E = \{ab, ac, cd, bd\}$

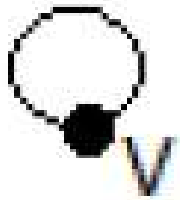


$V = \{\dots\dots\dots\}$

$E = \{\dots\dots\dots\}$

Loop

In a graph, if an edge is drawn from vertex to itself, it is called a loop.



$$V=\{v\}$$

$$E=\{vv\}$$

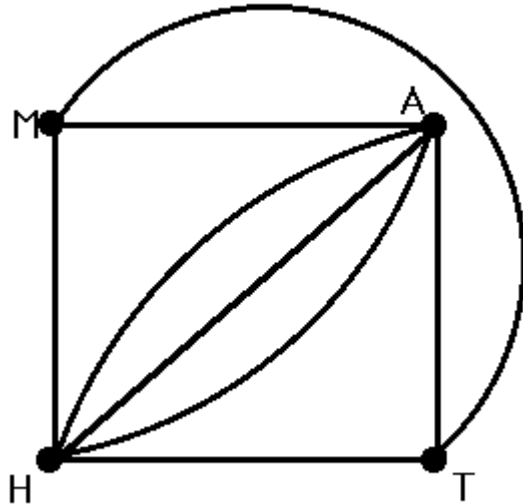


$$V=\{ \quad \}$$

$$E=\{ \quad \}$$

Degree of a Vertex

The number of edges incident to a vertex v is called the degree of the vertex and is denoted by $\deg(v)$. Equivalently, the degree of a vertex is the number of vertices adjacent to it.



Vertex	Degree
M	3
A	5
T	3
H	5

Pendent Vertex

By using degree of a vertex, we have a two special types of vertices. A vertex with degree one is called a pendent vertex.



Isolated Vertex

A vertex with degree zero is called an isolated vertex.

A diagram of an isolated vertex labeled 'a'. It consists of a single black dot with the letter 'a' centered below it. The entire diagram is enclosed in a light blue rounded rectangle.

a

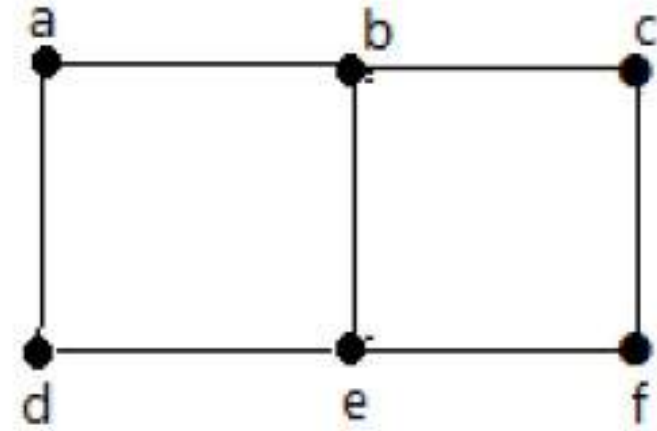
A diagram of an isolated vertex labeled 'b'. It consists of a single black dot with the letter 'b' centered below it. The entire diagram is enclosed in a light blue rounded rectangle.

b

Adjacency

In a graph, two vertices are said to be **adjacent**, if there is an edge between the two vertices. Here, the adjacency of vertices is maintained by the single edge that is connecting those two vertices.

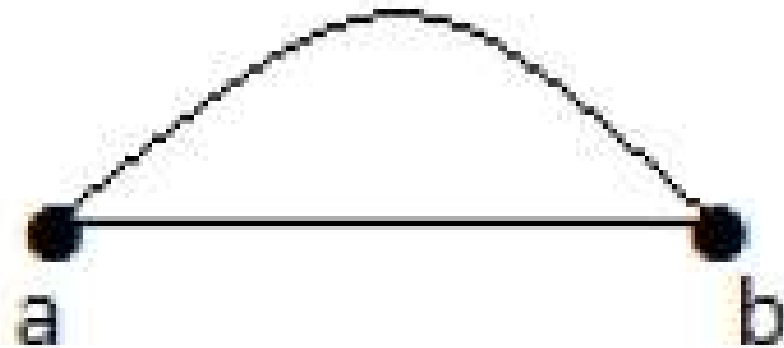
In a graph, two edges are said to be adjacent, if there is a common vertex between the two edges. Here, the adjacency of edges is maintained by the single vertex that is connecting two edges.



- In the above image, 'a' and 'b' are the adjacent vertices, as there is a common edge 'ab' between them.
- 'ab', 'bc' and 'be' are the adjacent edges, as there is a common vertex 'b' between them.

Parallel Edges

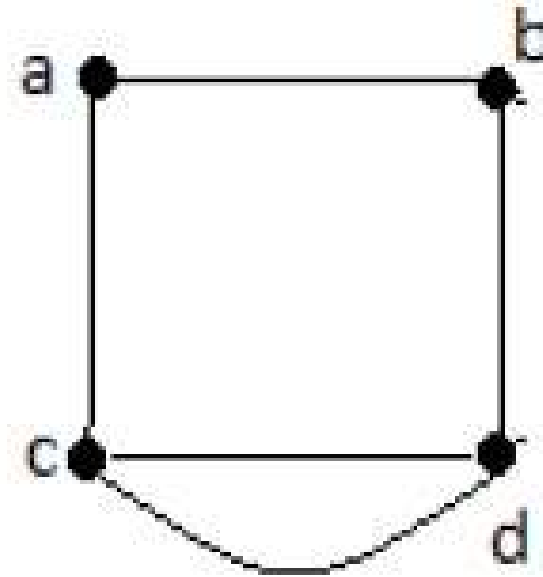
In a graph, if a pair of vertices is connected by more than one edge, then those edges are called parallel edges.



$V = \{a, b\}$
 $E = \{ab, ab\}$

Multi Graph

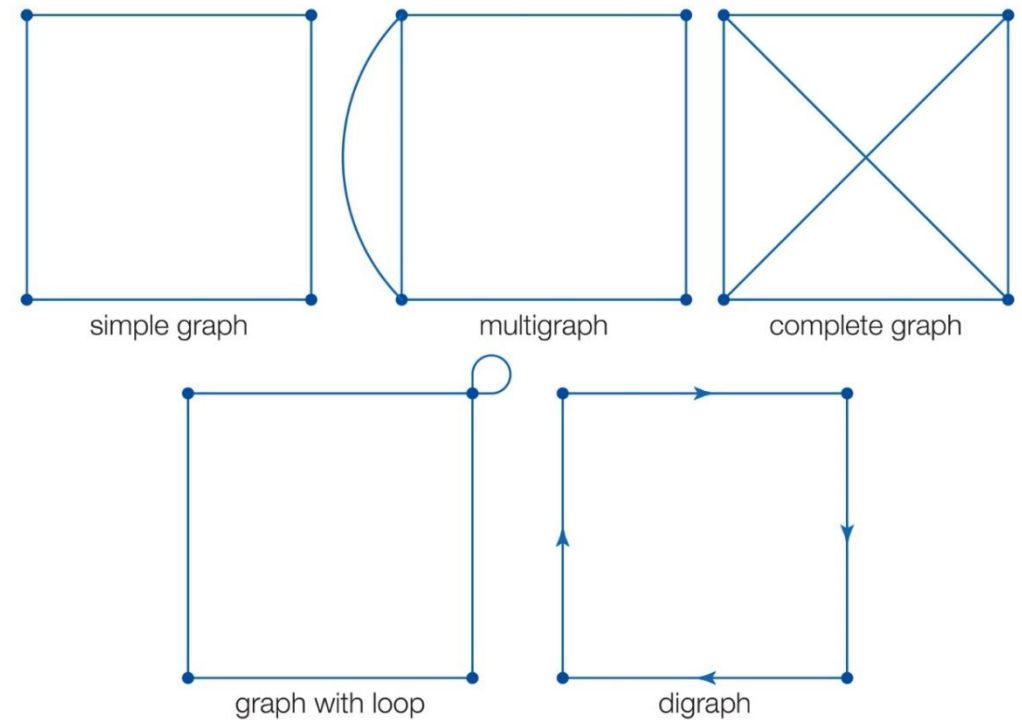
A graph having parallel edges is known as a Multigraph.



In the above graph, there are five edges 'ab', 'ac', 'cd', 'cd', and 'bd'. Since 'c' and 'd' have two parallel edges between them, it is a Multigraph.

Complete Graph, Digraph, and Simple Graph

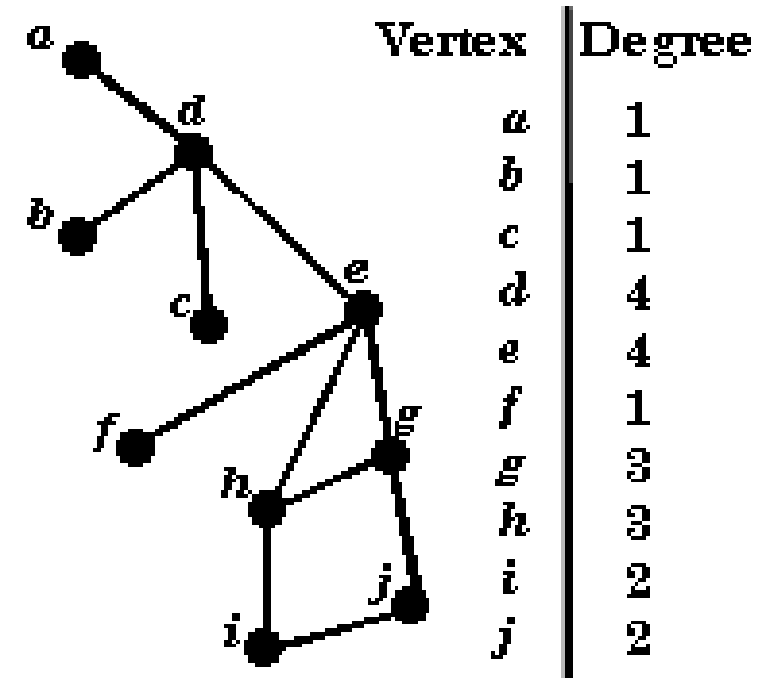
- A **simple graph**, also called a strict graph, is an unweighted, undirected graph containing no **graph** loops or multiple edges
- A **complete graph** is a graph in which every vertex is connected to every other vertex by an edge. If a complete graph has n vertices, then each vertex has degree $n - 1$.
- A **directed graph** (or **digraph**) is a graph that is made up of a set of vertices connected by edges, where the edges have a direction associated with them.
- A **complete digraph** is a directed graph in which every pair of distinct vertices is connected by a pair of unique edges.



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Degree of Vertex

- It is the number of vertices adjacent to a vertex V .
- **Notation** – $\deg(V)$.
- A vertex can form an edge with all other vertices except by itself. So the degree of a vertex will be up to the **number of vertices in the graph minus 1 for a simple graph**.
- If there is a loop at any of the vertices, then it is not a Simple Graph.



Degree of Vertex in an Undirected Graph

An undirected graph has no directed edges. Consider the following examples.

In this Undirected Graph,

$\deg(a) = 2$, as there are 2 edges meeting at vertex 'a'.

$\deg(b) = 3$, as there are 3 edges meeting at vertex 'b'.

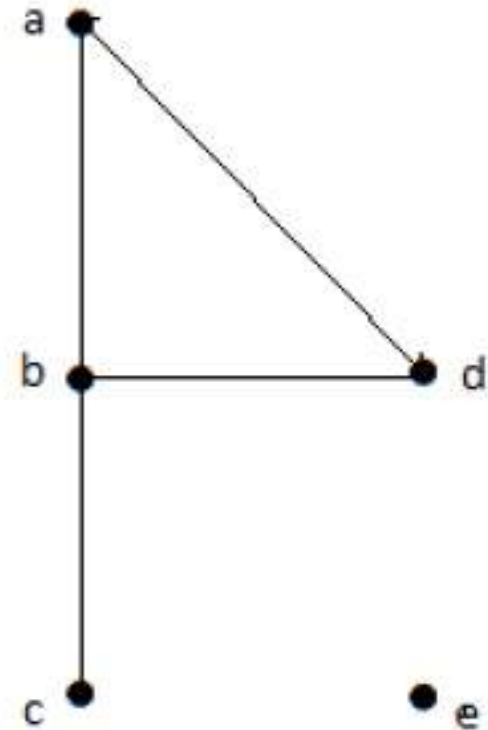
$\deg(c) = 1$, as there is 1 edge formed at vertex 'c'

So 'c' is a **pendent vertex**.

$\deg(d) = 2$, as there are 2 edges meeting at vertex 'd'.

$\deg(e) = 0$, as there are 0 edges formed at vertex 'e'.

So 'e' is an **isolated vertex**.



Degree of Vertex in an Undirected Graph

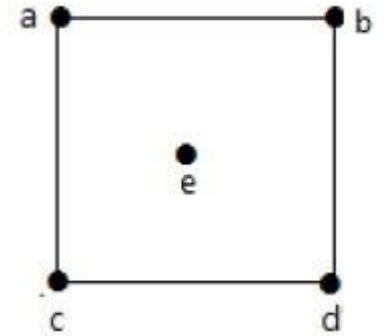
$\deg(a) = ,$

$\deg(b) = ,$

$\deg(c) = ,$

$\deg(d) = ,$

$\deg(e) =$



Degree of Vertex in a Directed Graph

In a directed graph, each vertex has an **indegree** and an **outdegree**.

Indegree of a Graph

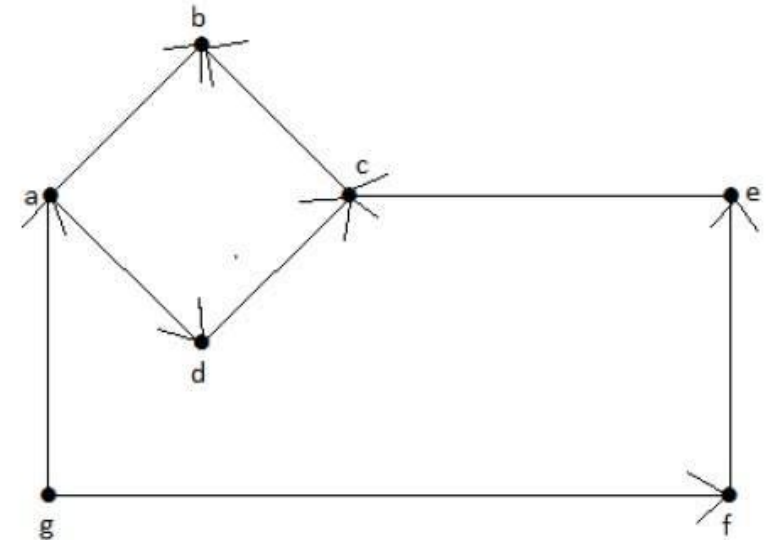
Indegree of vertex V is the number of edges which are coming into the vertex V .

Notation – $\deg^-(V)$.

Outdegree of a Graph

Outdegree of vertex V is the number of edges which are going out from the vertex V .

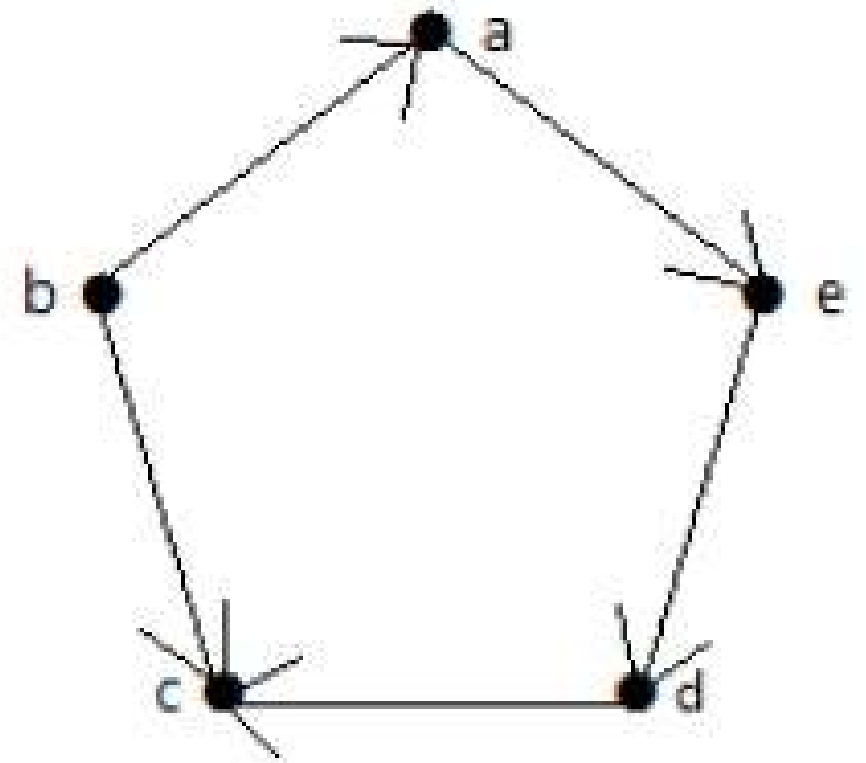
Notation – $\deg^+(V)$.



Vertex	Indegree	Outdegree
a	1	2
b	2	0
c	2	1
d	1	1
e	1	1
f	1	1
g	0	2

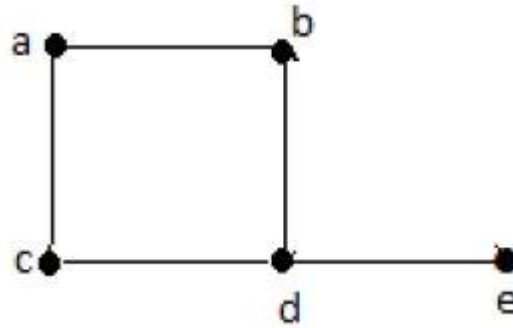
Degree of Vertex in a Directed Graph

Vertex	Indegree	Outdegree
a	1	1
b	0	2
c	2	0
d	1	1
e	1	1



Degree Sequence of a Graph

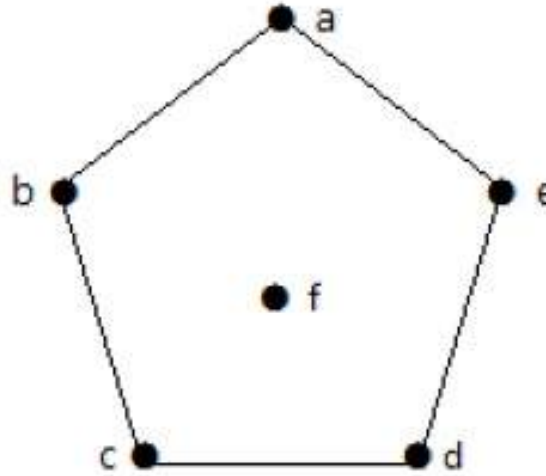
If the degrees of all vertices in a graph are arranged in descending or ascending order, then the sequence obtained is known as the degree sequence of the graph.



Vertex	A	b	c	d	e
Connecting to	b,c	a,d	a,d	c,b,e	d
Degree	2	2	2	3	1

In the above graph, for the vertices {d, a, b, c, e}, the degree sequence is {3, 2, 2, 2, 1}.

Degree Sequence of a Graph



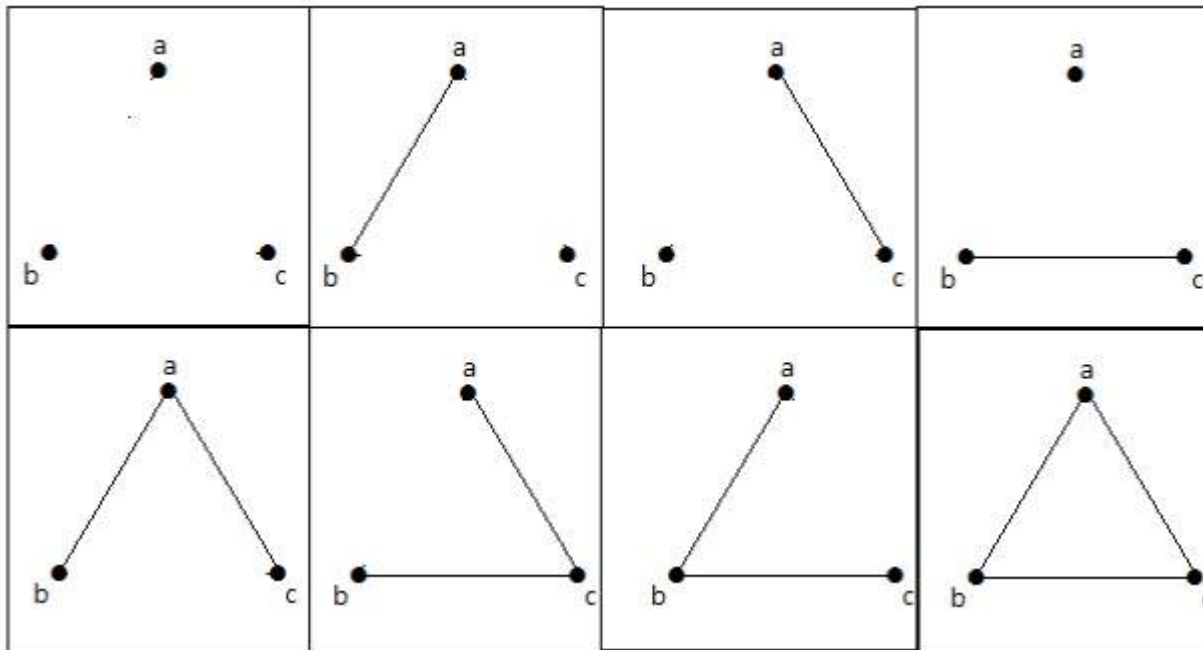
Vertex	A	b	c	d	e	f
Connecting to	b,e	a,c	b,d	c,e	a,d	-
Degree	2	2	2	2	2	0

In the above graph, for the vertices $\{a, b, c, d, e, f\}$, the degree sequence is $\{2, 2, 2, 2, 2, 0\}$.

Relationship between Edges, Vertices, and Simple Graphs

Ex.

1. Find the maximum number of edges with $n=3$ vertices
2. maximum number of simple graphs with $n=3$ vertices



Relationship between edges, vertices, and Simple Graphs

□ The maximum number of edges possible in a simple graph with 'n' vertices is nC_2

where ${}^nC_2 = n(n - 1)/2$.

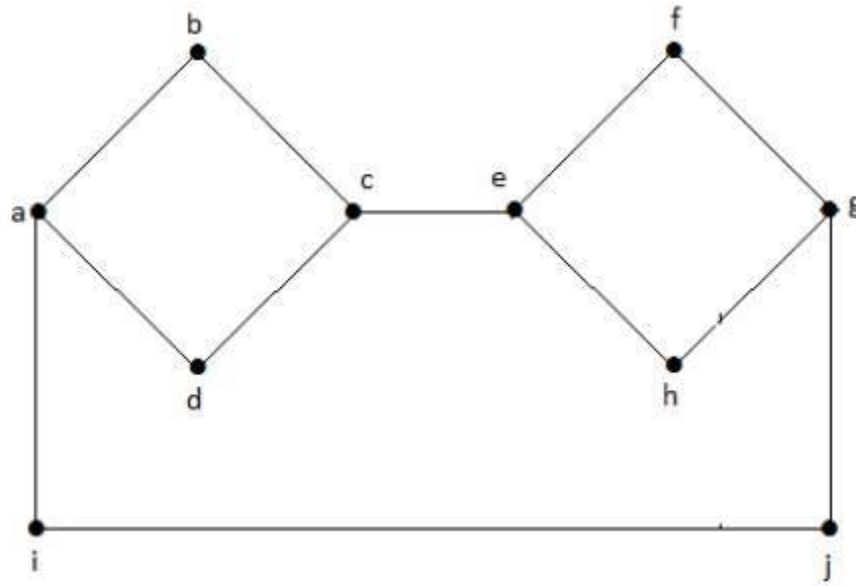
□ The number of simple graphs possible with 'n' vertices = $2^{nc_2} = 2^{n(n-1)/2}$

Ex.

1. Find the maximum number of edges with $n=4$ vertices
2. maximum number of simple graphs with $n=4$ vertices

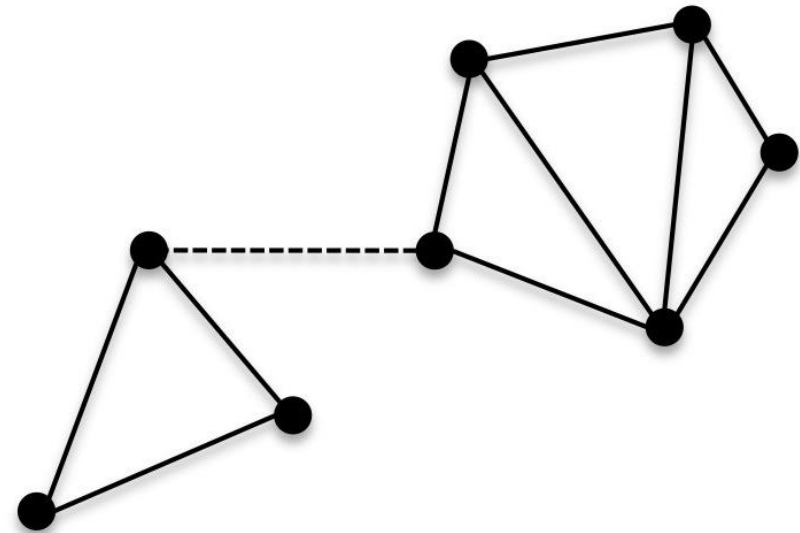
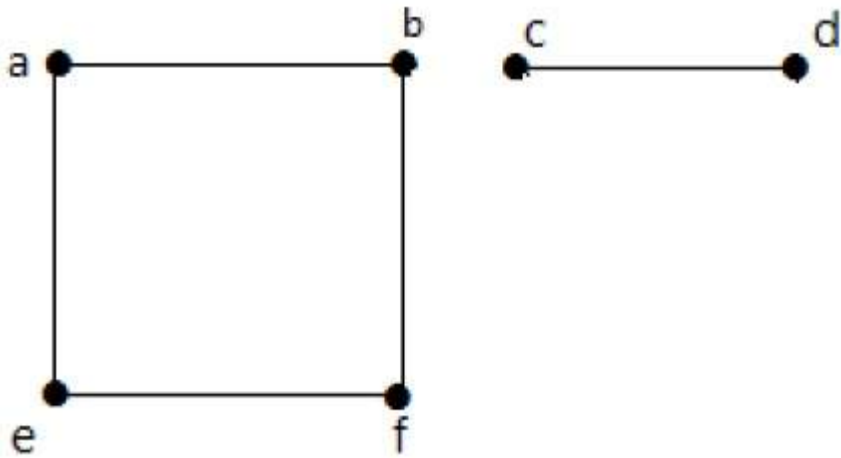
Connected Graph

A graph G is said to be connected **if there exists a path between every pair of vertices**. There should be at least one edge for every vertex in the graph. So that we can say that it is connected to some other vertex at the other side of the edge.



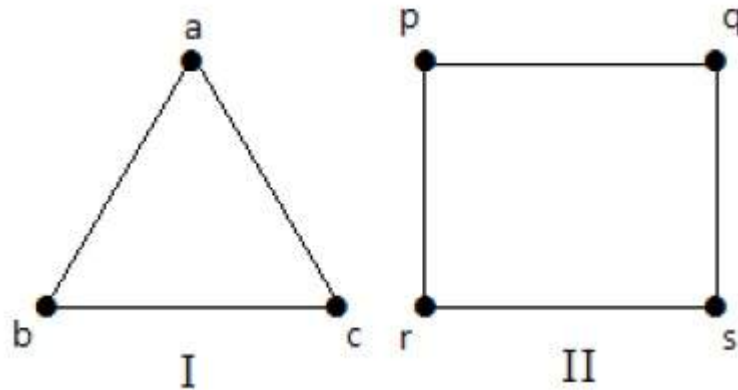
Disconnected Graph

A graph that is not connected is **disconnected**. An undirected graph G is said to be **disconnected** if there exist two nodes in G such that no path in G has those nodes as endpoints.



Regular Graph

A graph G is said to be regular, **if all its vertices have the same degree**. In a graph, if the degree of each vertex is ' k ', then the graph is called a ' k -regular graph'.

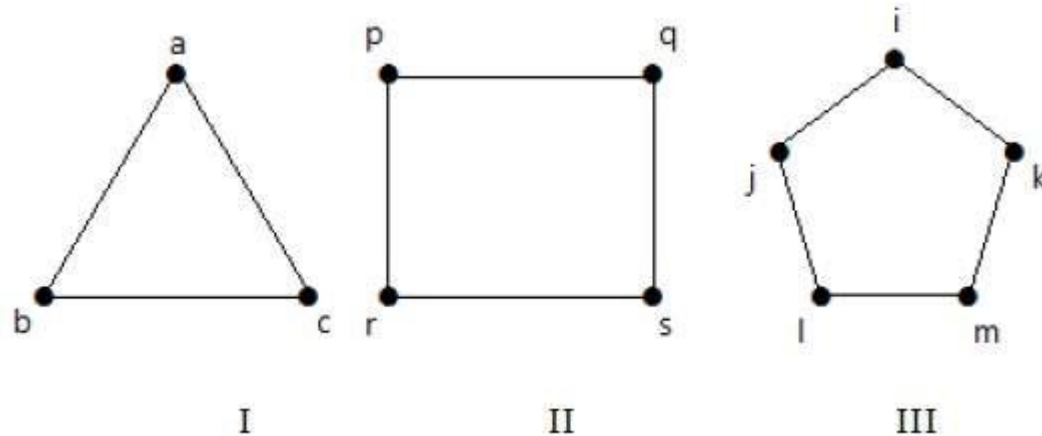


Eg. all the vertices have degree 2. They are called 2-Regular Graphs.

Cycle Graph

A simple graph with ' n ' vertices ($n \geq 3$) and ' n ' edges is called a cycle graph if all its edges form a cycle of length ' n '. If the **degree of each vertex in the graph is two**, then it is called a Cycle Graph.

Notation – C_n



Graph I has 3 vertices with 3 edges which is forming a cycle 'ab-bc-ca'.

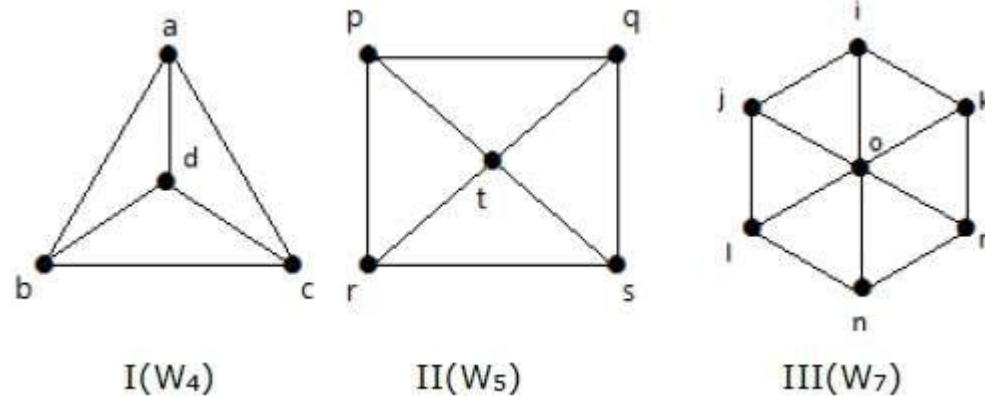
Graph II has 4 vertices with 4 edges which is forming a cycle 'pq-qs-sr-rp'.

Graph III has 5 vertices with 5 edges which is forming a cycle 'i-j-k-l-m-i'.

Wheel Graph

A wheel graph is obtained from a cycle graph C_{n-1} by adding a new vertex. That new vertex is called a **Hub** which is connected to all the vertices of C_n .

Notation – W_n

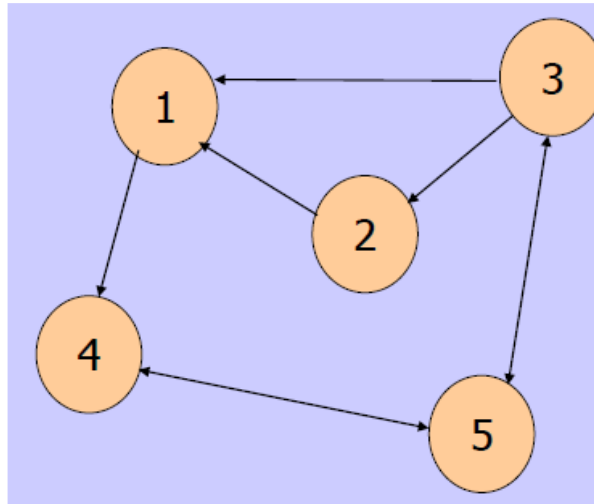


Ex. Find the number of edges in Wheel Graphs I, II, and III

No. of edges in W_n = No. of edges from hub to all other vertices +
No. of edges from all other nodes in cycle graph without a hub.
= $(n-1) + (n-1)$
= $2(n-1)$

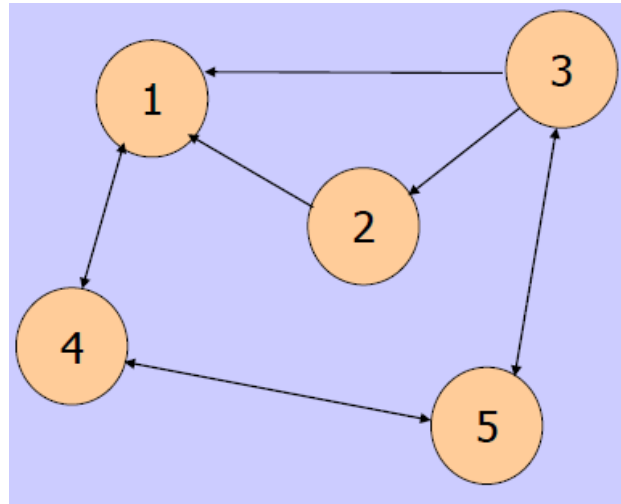
Circuit

A path whose first and last vertices are the same and no edge is repeated then the path is called a circuit. The path 3,2,1,4,5,3 is a circuit.



Cycle

A **circuit** that doesn't repeat vertices is called a **cycle**.

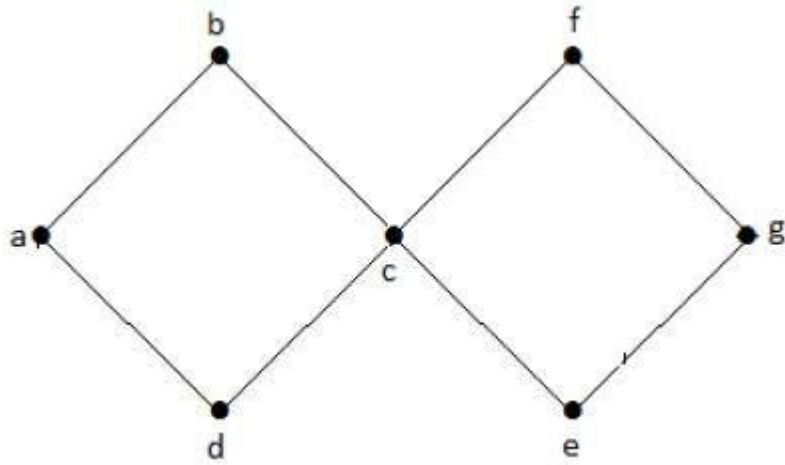


1,4,5,3,1 is a cycle

1,4,5,4,1 is not a cycle

Cyclic Graph

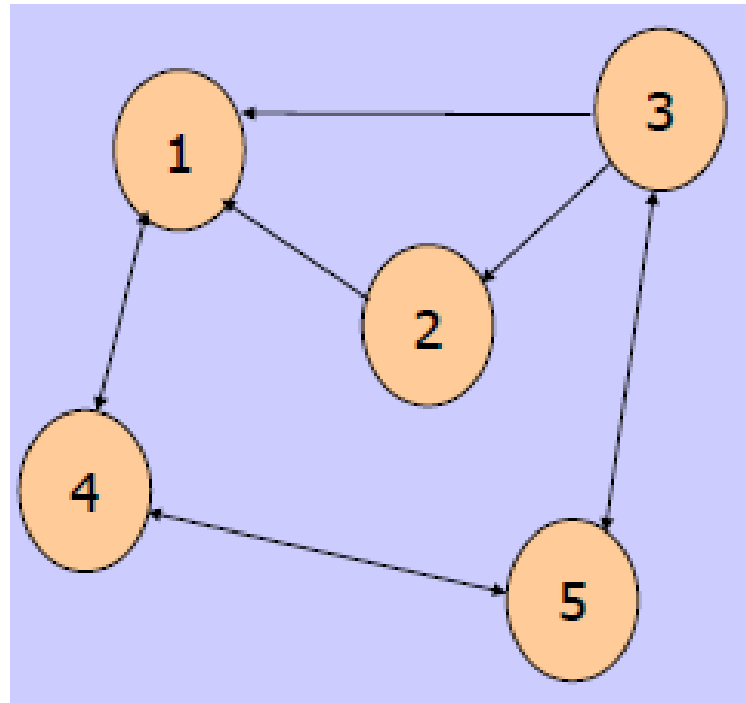
A graph **with at least one** cycle is called a cyclic graph.



In the above example graph, we have two cycles a-b-c-d-a and c-f-g-e-c. Hence it is called a cyclic graph.

Hamiltonian Cycle

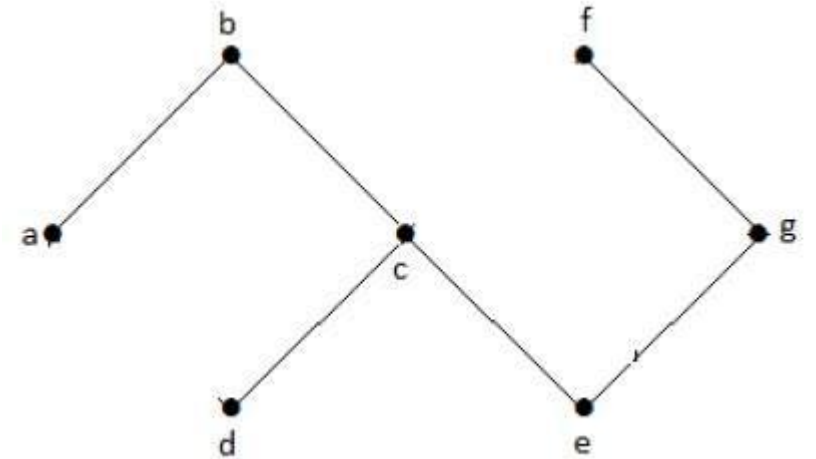
A Cycle that contains all the vertices of the graph



1,4,5,3,2,1 is a Hamiltonian Cycle

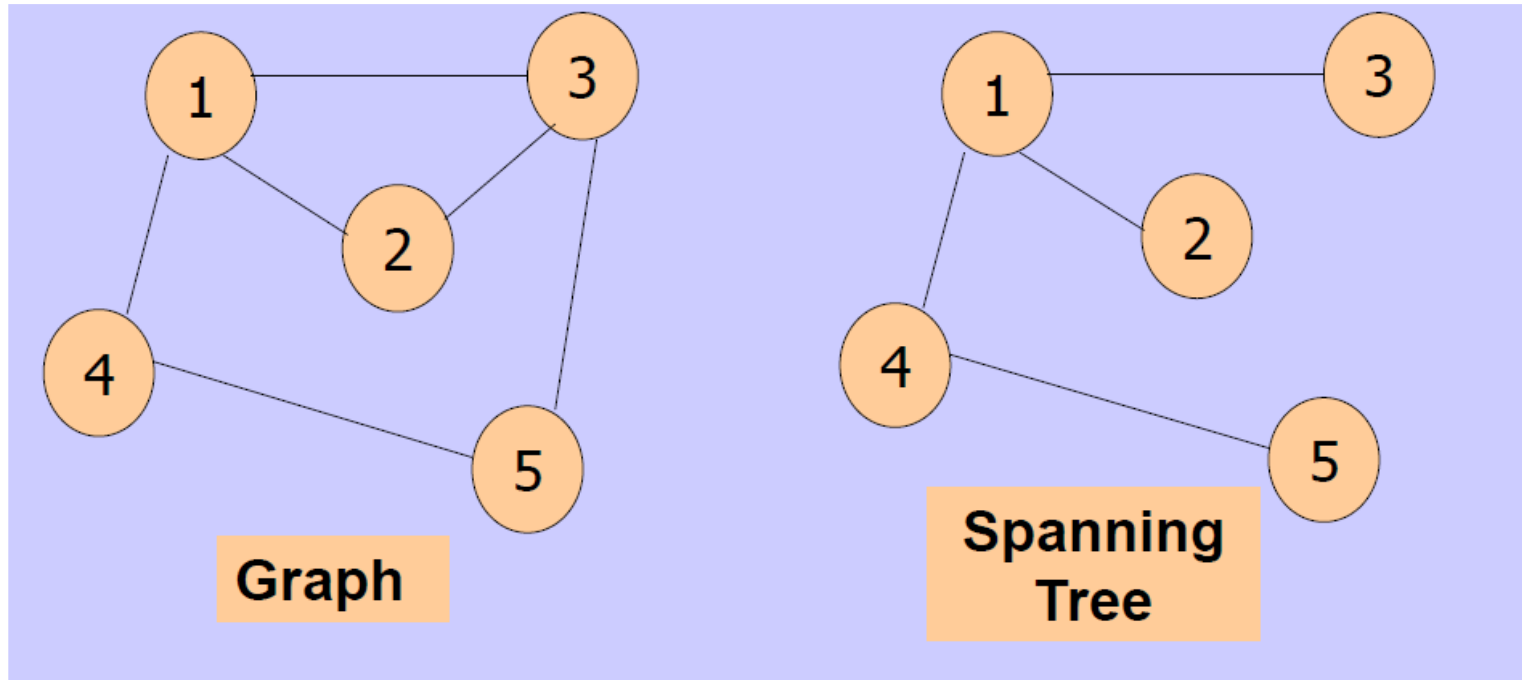
Trees

- Connected and contains no cycles.
- Has $V-1$ number of edges.
- Any two vertices of a Tree are connected by exactly one path.
- Tree contains no cycles, but the addition of any new edge creates exactly one cycles.
- Trees can only represent relations of a hierarchical type.
- Graph is a data structure in which this limitation is lifted.
- A graph need not be a tree but a tree must be a graph.



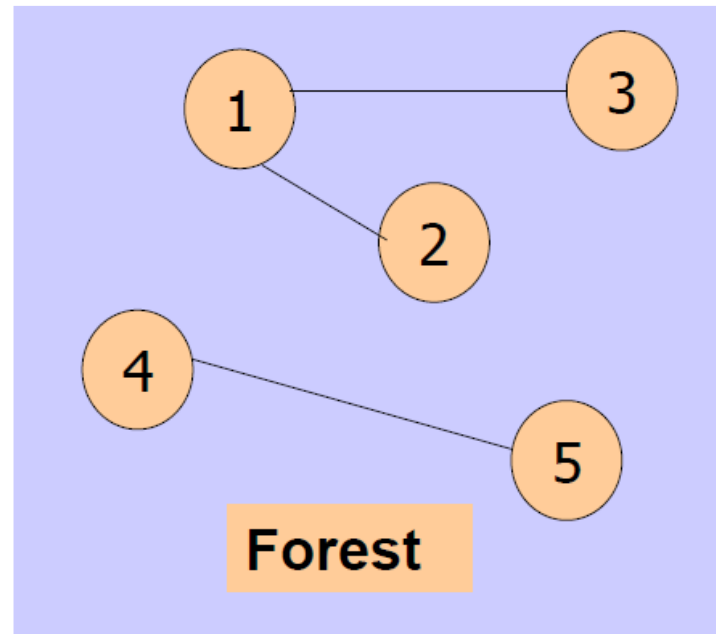
Spanning Tree

The *Spanning Tree* of a Graph G is a sub graph of G that is a tree and contains all the vertices of G .



Forest

Disjoint union of trees.



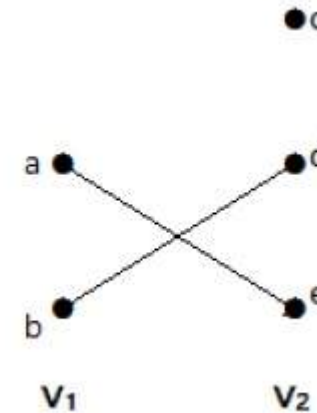
Bipartite Graph

A simple graph $G = (V, E)$ with vertex partition $V = \{V_1, V_2\}$ is called a bipartite graph **if every edge of E joins a vertex in V_1 to a vertex in V_2 .**

In general, a Bipartite graph has two sets of vertices, let us say, V_1 and V_2 , and if an edge is drawn, it should connect any vertex in set V_1 to any vertex in set V_2 .

In this graph, you can observe two sets of vertices – V_1 and V_2 .

Here, two edges named 'ae' and 'bd' are connecting the vertices of two sets V_1 and V_2 .

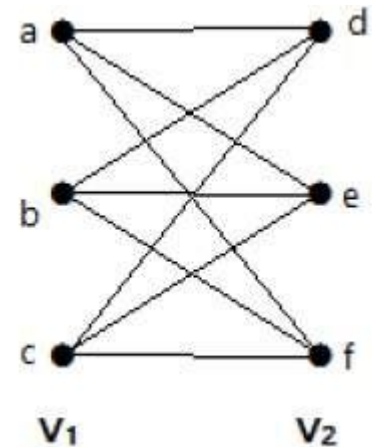


Complete Bipartite Graph

A bipartite graph 'G', $G = (V, E)$ with partition $V = \{V_1, V_2\}$ is said to be a complete bipartite graph if every vertex in V_1 is connected to every vertex of V_2 .

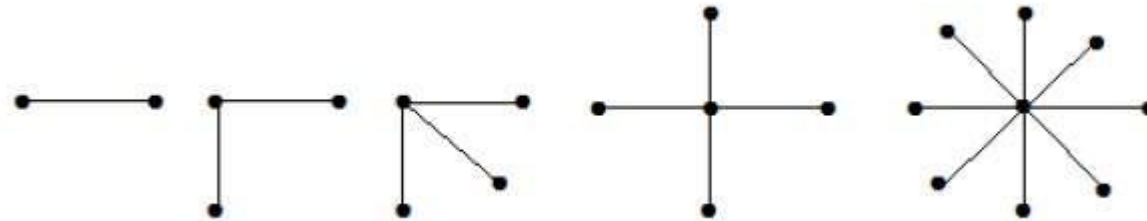
In general, a complete bipartite graph connects each vertex from set V_1 to each vertex from set V_2 .

The following graph is a complete bipartite graph because it has edges connecting each vertex from set V_1 to each vertex from set V_2 .



Star Graph

A star graph is a complete bipartite graph if a single vertex belongs to one set and all the remaining vertices belong to the other set.



In the above graphs, out of 'n' vertices, all the 'n-1' vertices are connected to a single vertex. Hence it is in the form of $K_{1, n-1}$ which are star graphs.

Famous Problems on Graphs

- The Euler cycle (or tour) problem: Is it possible to traverse each of the edges of a graph exactly once, starting and ending at the same vertex?
- The Hamiltonian cycle problem: Is it possible to traverse each of the vertices of a graph exactly once, starting and ending at the same vertex?
- The traveler salesman problem: Find the shortest path in a graph that visits each vertex at least once, starting and ending at the same vertex?
- The planar graph: Is it possible to draw the edges of a graph in such a way that edges do not cross?

Thank You!!
